

Regulatory Risk and Rate Volatility in M&V: Lessons from California's Solar Market

Background: California's Midday Solar Situation

California now generates roughly 32% of its annual electricity from solar, and during midday hours (8 a.m.–4 p.m.) solar supplies nearly half of CAISO grid demand. Solar capacity during peak spring months grew from 10.2 GW in 2020 to 18.8 GW in 2025. The grid regularly produces more solar than it can consume, resulting in curtailment and negative wholesale prices.

Battery storage has become essential to managing this glut — California's battery capacity grew nearly twentyfold from 0.6 GW in 2020 to 11.7 GW in 2024, absorbing midday surplus and discharging during the critical 5–9 p.m. evening ramp.

The Wholesale/Retail Disconnect

One of the most important market dynamics for M&V professionals to understand is the near-complete decoupling of wholesale and retail electricity prices in California:

- **Wholesale midday price:** ~3–5 cents/kWh (often negative during solar peaks)
- **Average retail residential price:** ~28 cents/kWh (December 2025)

Retail rates are set through CPUC regulatory proceedings, not real-time markets. They are dominated by fixed cost recovery — wildfire hardening, infrastructure investment, utility debt service, and cross-subsidies — that does not respond to wholesale conditions.

Implication: A saved kWh at retail is still worth ~28 cents to a California commercial or institutional client (school district, university, government facility), regardless of what CAISO wholesale prices are doing. Efficiency remains highly valuable from the retail customer's perspective.

M&V Implications: Is Efficiency Still Valuable?

Yes — but with important nuance that students must internalize:

1. Avoided cost is now time-dependent

A kWh saved at noon during a negative-price event has near-zero wholesale value. A kWh saved at 7 p.m. on a hot evening is extremely high-value, displacing gas peakers. The **temporal dimension of the counterfactual now matters** — not just how much is saved, but *when*.

This maps directly to IPMVP's boundary and duration choices: M&V plans must account for the timing of savings, not just the aggregate.

2. The retail/wholesale stack determines whose perspective you're in

- A school district or government facility pays retail rates — efficiency savings are as valuable as ever.
- A large industrial customer with wholesale market exposure faces an entirely different calculus.
- Students must identify **which avoided cost stack** applies to their client before designing an M&V protocol.

3. Efficiency shifts the duck curve

Demand reduction during the 5–9 p.m. evening ramp is now arguably the **highest-value efficiency outcome** in California. Load shifted away from evening peaks reduces pressure on batteries and gas peakers. M&V protocols should track *when* savings occur, not just aggregate them annually — hourly or interval data is increasingly necessary for rigorous analysis.

4. The counterfactual baseline is destabilized

If retail rates keep rising while wholesale collapses at midday, what is the correct avoided cost for a savings calculation? The competent M&V professional must:

- Model rate structures explicitly (flat, TOU, demand charges, fixed charges)
- Account for the client's actual tariff, not a generic \$/kWh
- Acknowledge rate uncertainty as a documented assumption

Planning for Future Rate Volatility: Regulatory Risk

The fundamental problem

California retail rates are not forecastable in the conventional sense — they are **political and regulatory outcomes**, not commodity prices. The ~8% compound annual growth rate since 2013 is driven by PG&E's bankruptcy settlement, wildfire liability, grid hardening mandates, and fixed cost spreading as efficient customers reduce consumption. This cannot be hedged the same way as commodity risk.

Strategies for M&V professionals and their clients

Avoided cost stacking, not single-rate assumptions

Build savings calculations that identify *which cost components* the efficiency measure actually avoids:

- Energy (kWh) charges — subject to rate changes
- Demand (kW) charges — often more stable but restructured over time
- Fixed monthly charges — increasingly large, and *not* reduced by efficiency
- Transmission and distribution components

Knowing which components are displaced makes savings estimates more defensible across rate changes and more honest with clients.

Scenario banding instead of point estimates

Rather than a single escalation rate, present a range:

- **Conservative:** Current rate, no escalation
- **Base:** Historical CAGR (~6–8% for California IOUs)
- **Stress:** Rate restructuring that shifts more cost to fixed charges, reducing per-kWh savings

This is an application of the epistemological humility central to competent counterfactuals: we know rates will change; we don't know how. The honest M&V professional quantifies the range rather than hiding it in a single assumption.

TOU alignment as risk management

As California moves more customers to time-of-use rates, efficiency measures that reduce **evening peak loads** become structurally more defensible. Measures that only reduce midday consumption face the risk that future rate design will value those hours even less. Evaluate measures partly on whether savings accrue during rate-stable versus rate-volatile hours.

Fixed charge exposure

This is underappreciated. California's income-graduated fixed charge (rolled out 2024) moves cost recovery away from per-kWh charges. A client who cuts consumption significantly after an efficiency retrofit may find bill savings smaller than projected if a larger share of cost has migrated to fixed charges. The **counterfactual bill** — not just counterfactual energy use — is the real target, and these can diverge significantly.

Contractual structures that allocate rate risk explicitly

In ESPCs and guaranteed savings contracts, savings are typically expressed in **physical units** (kWh, therms) for this reason — locking in dollar savings against an uncertain future rate is extremely difficult. The ESCO takes performance risk; the client often retains rate risk implicitly. M&V professionals should be able to articulate where that boundary sits and advise clients on whether they understand what risk they are holding.

Core Teaching Point

This material connects directly to the foundational principle of competent counterfactuals:

A counterfactual is not discovered — it is designed.

The avoided cost assumption embedded in any savings calculation is itself a designed counterfactual. Regulatory risk is, in effect, a **non-routine adjustment waiting to happen**. The competent M&V professional:

1. Makes the rate assumption explicit and documents its basis
2. Presents scenario ranges rather than false precision
3. Flags the conditions under which the assumption would need revision
4. Aligns savings timing with the rate structure the client actually faces

Students who internalize this will be far more credible in front of sophisticated clients — and far more defensible if their savings claims are ever challenged.

Developed from classroom discussion on California solar markets, wholesale/retail price dynamics, and regulatory risk planning. Cross-reference: IPMVP boundary and duration decisions; non-routine adjustments; performance contracting risk allocation.

